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Microsoft Fabric COMMUNITY CONFERENCE | Microsoft SQL COMMUNITY CONFERENCE

ATLANTA MARCH 16 - 20, 2026

Microsoft Fabric Capacities

Essentials for Smarter Operations

Sanjay Raut, Pat Mahoney, Matt Stenzel

Principal PM's - Azure Data CAT

Meet the Speakers and Proctors

- Speakers
  - Matt Stenzel
  - Sanjay Raut
  - Pat Mahoney

- Proctors
  - Tim Bindas
  - Pratik Patel

Full-Day Workshop Agenda Timeline

9:00 AM

Session 1 - Capacity Concepts (75 min)

CU model, bursting, throttling, hands-on lab [1]

10:30 AM

Session 2 - Recovery & Advanced Config (90 min)

Throttling recovery, advanced settings, licensing [1]

12:45 PM

Session 3 - Monitoring & Events (150 min)

Metrics App, Real-Time Hub, chargeback demos [1]

3:30 PM

Session 4 - Planning & Optimization (30 min)

Best practices, scale decisions, workload isolation [1]

4:00 PM

Session 5 - Q&A with Leadership (30 min)

Open forum with Capacity Engineering Leadership [1]

- Structured learning progression: From foundational concepts to advanced topics
- Interactive engagement with hands-on labs, live demonstrations,
- Bonus - dedicated Q&A sessions towards end of the day to ask further clarifications around real capacity scenarios
- Coffee Breaks:
  - 10:15 AM to 10:30 AM
  - 3:15 PM to 3:30 PM
- Lunch Break:
  - 12:00 Noon to 12:45 PM

Session 4 – Capacity Planning & Optimization

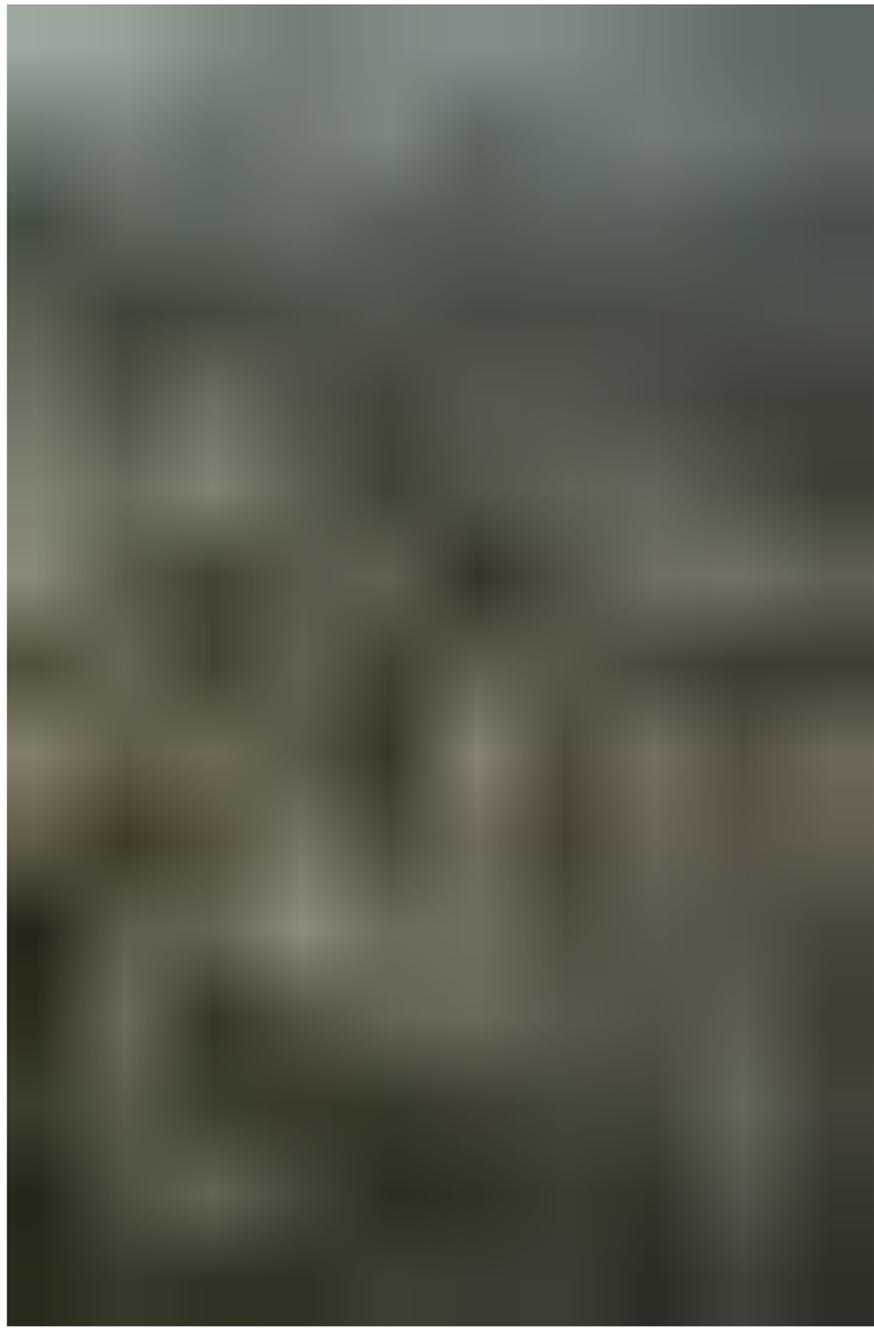
Reframing Capacity Planning

Reframing Capacity Planning

- **Context and problem**
- Capacity planning treated as one-time SKU sizing
- Leads to **reactive scale-ups**, **throttling surprises**, and cost inefficiencies

- **Core idea**
- Continuous system design should span proof of concept, self-service, and enterprise stages.
- Planning must account for variability and concurrency, not only peak demand,
- operating within sustainable envelopes that combine bursting and smoothing.

Capacity planning means building a resilient, cost-aware compute system that **evolves alongside adoption**



Capacity Planning Challenge

The Core Tension

Fabric F-SKUs are **fixed compute pools**

- Scaling is a deliberate, manual decision.
- No real-time auto-scale today (except few)

The Stakes

- **Unplanned surges** → throttled queries, degraded UX
- **Poor sizing** → blown budgets or SLA violations

Deliberate planning is **non-negotiable**.

Right-size ahead of demand — *bursting and smoothing absorb short-term peaks, **not sustained overloads.***

Tiered Capacity Strategy – SLA Driven

EXPLORATORY (GENERAL LANE)

Treating all workloads equally increases risk by forcing mission-critical services to compete with experiments.

SELF-SERVICE (SHARED)

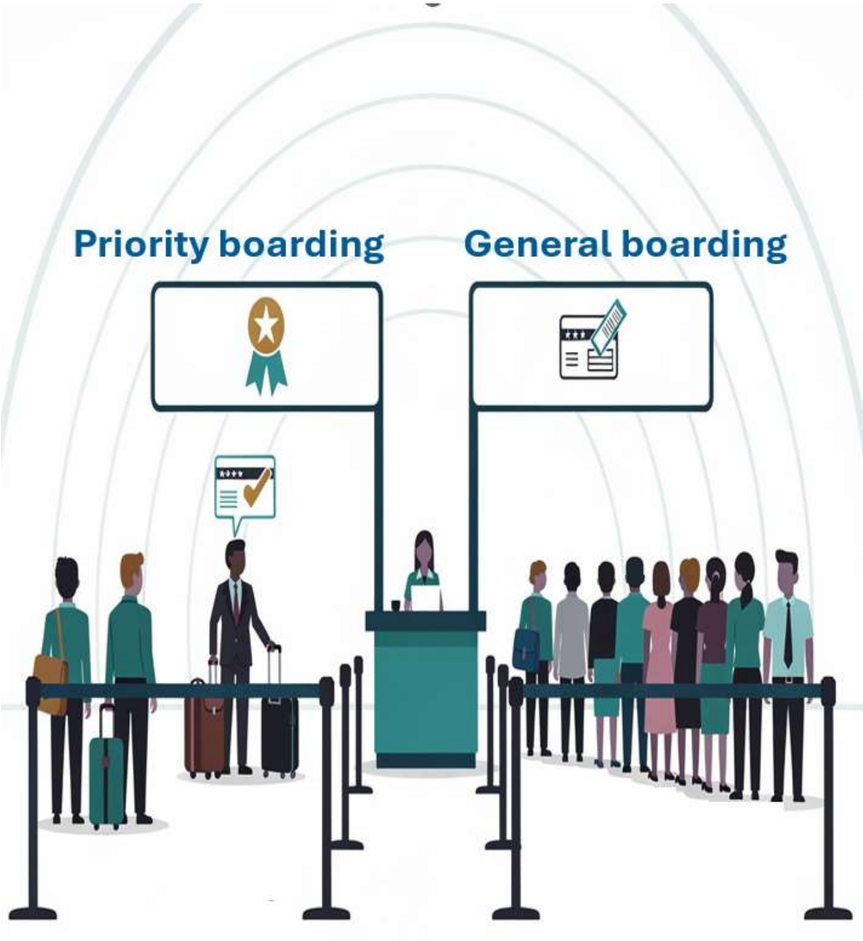
Differentiated capacity enables experimentation and innovation without disrupting protected environments.

ENTERPRISE-MANAGED (PROTECTED)

Tiering by criticality protects mission-critical workloads with clear ownership, controls, and SLA commitments.

GOVERNANCE & PLACEMENT

Governance boundaries should align to the adoption roadmap so teams scale responsibly as workloads mature.



Match capacity placement to workload criticality - **SLA driven**

Monitoring Signals and Decision Inputs

- Key Tools
  - **Fabric Capacity Metrics App** — primary utilization tracking
  - **RTH for Capacity Events**— alerts and trend analysis over longer period (> 30 days)
  - **FUAM** and other tools

- Action Thresholds
  - Watch for: rising 24-hr utilization,
  - frequent rejections,
  - persistent capacity debt.

> 80% avg CU for 60+ min

Scale out or split workloads

< 30% peak CU sustained

Scale down or consolidate

Scale Up, Out, Or Optimize

01

Optimize First

Address inefficiencies by fixing reports, refresh patterns, and pipelines, and optionally use a Metrics App walk-through to identify top consumers.

02

Scale Up for Growth

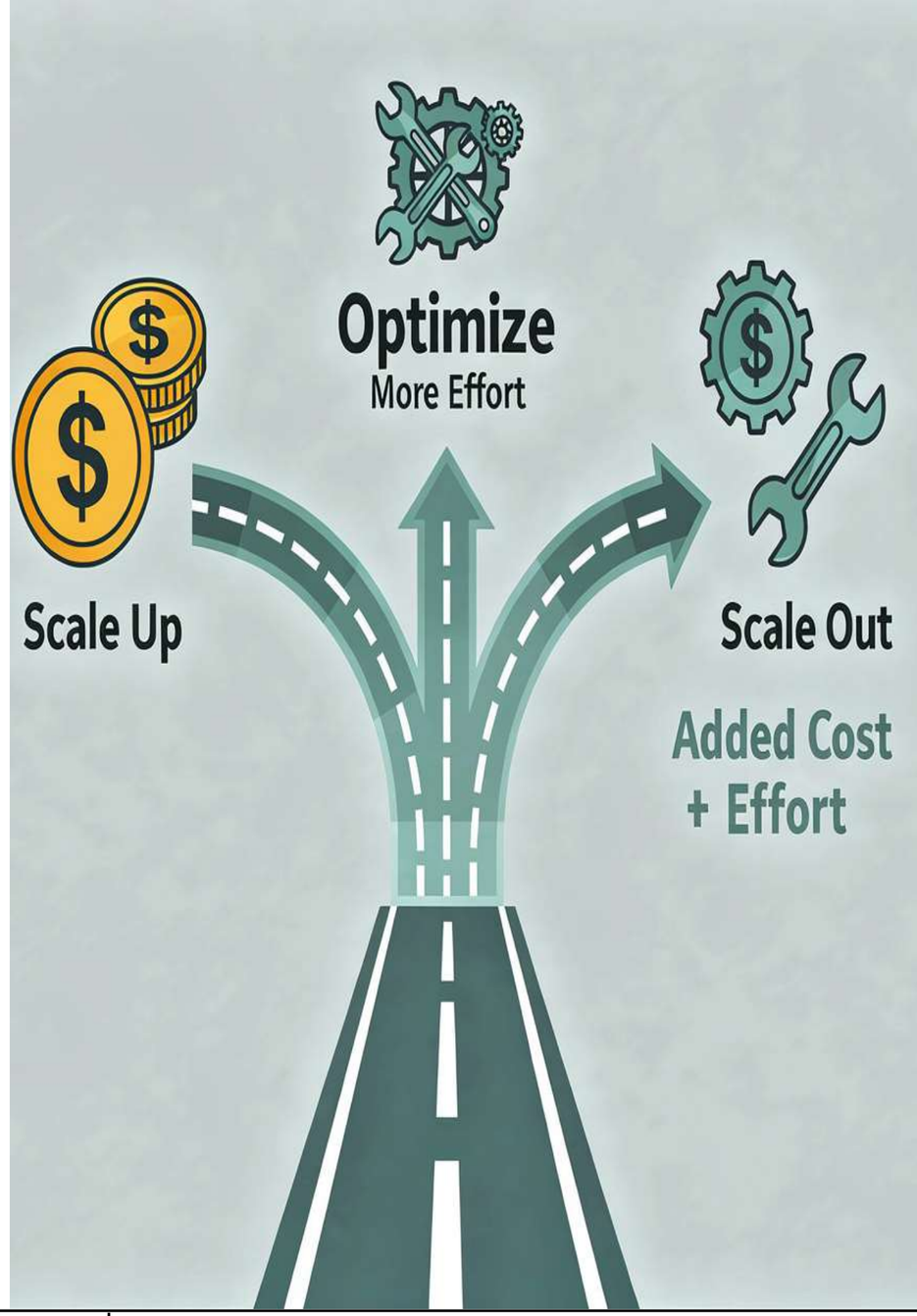
Increase SKU capacity when growth is predictable and sustained, recognizing the direct impact on cost efficiency and overall governance posture.

03

Scale Out for Isolation

Add dedicated capacity when contention exists or workload isolation is required to improve workload separation, governance, and operational resilience.

1) Optimize (first): **fix inefficiencies -** reports, refresh patterns, pipelines
2) Scale up: for sustained, **predictable** growth
3) Scale out: **isolation** for mission-critical / noisy neighbor risk



Surge Protection: Safeguarding Capacity Health

Capacity-Level Controls

- Custom background rejection thresholds (e.g., block at **70% smoothing**)
- Auto-resume once usage drops below recovery threshold (e.g., 50%)

Workspace-Level Controls

- Max daily CU % per workspace — exceeding it **blocks that workspace**
- Mark critical workspaces as **mission-critical** — exempt from auto-blocking (CxO Reporting)

Surge protection is a **safety net**, not a sizing substitute. It shields interactive workloads when unexpected surges hit — enable it on every production capacity.

Governance and Continuous Optimization

Establish a COE

Center of Excellence owns **fair-use policies**, **chargeback/showback**, and capacity procurement decisions.

Cadenced Reviews

Monthly or **quarterly reviews** detect growth trends and cost-cutting opportunities before they become problems.

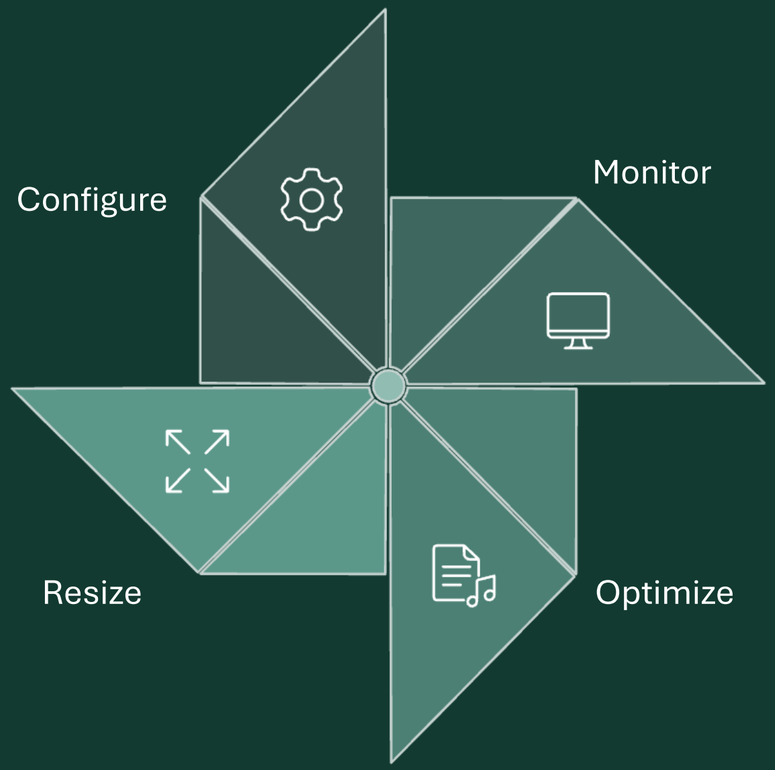
Proactive Scheduling

Pause or scale down off-peak. Scale up ahead of known events — quarter-end reports, major ETL windows.

Iterate the Plan

Revisit sizing, surge protection settings, and Autoscale decisions whenever usage patterns shift materially.

> Even appropriately sized capacity degrades without **ownership**, **governance** and **shared accountability**.



Key Takeaways

Right-Size Proactively

Size for sustained average load — bursting and smoothing handle short peaks.

Know Your Workloads

Interactive vs. background patterns drive sizing. Identify nightly ETL and report storm windows.

Monitor with Thresholds

Act at >80% avg CU. Don't wait for user complaints — set automated alerts.

Enable Surge Protection / Mission Critical

Circuit-breaker for capacities — blocks runaway workloads before a full throttle event.

Choose the Right Capacity Model

Hybrid strategy: reserved capacity for steady base load, Autoscale for irregular spikes.

Govern Continuously

COE + cadenced reviews + chargeback = a self-correcting, cost-predictable platform.

Fabric Capacity Planning Guide

https://aka.ms/FabricCapacityPlanning

Acknowledgment of Contributions

Core Contributors

Special thanks to Benni, Pat, Chris, Tim, Matthew, Ope, Lukasz, Edgar, Pankaj, Carlos, Matt, Sanjay, and others for their valuable presentations.

Workshop Inspiration

This workshop compiles insights from numerous team members' presentations shared at previous conferences, enriching the community's knowledge.

Gratitude and Support

The success of this workshop relies on the support and contributions of the CAT and Capacities Engineering Team.

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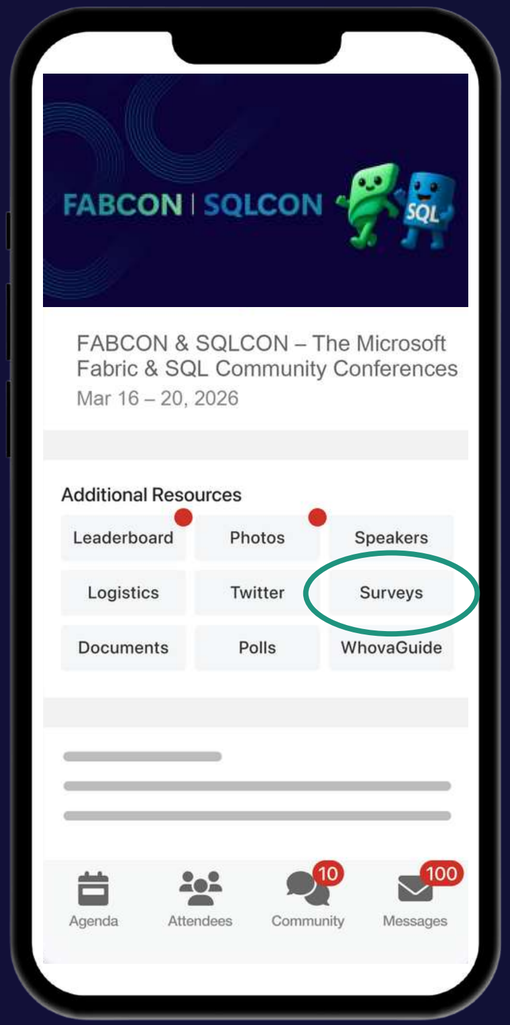

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